PROJECT REPORT

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NAME OF PROJECT: Bangladesh Indoor Air Quality Analysis

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1. Dataset Description

The dataset used in this project is the Residential Air Pollution in Bangladesh dataset. The notebook's own saved outputs show that the working file contains 10,000 observations and 22 features, recorded at 15-minute intervals across two seasons (winter and summer). This is confirmed independently by three separate cells: the raw dataframe display reports (10000 rows x 22 columns), the central_data table used for every mean/median/mode calculation reports (10000 rows x 15 columns), and the six categorical frequency tables in Section 4 all sum to 10,000.

The variables cover major indoor pollutants (PM2.5, PM10, CO2, CO, NO2 and TVOC), indoor and outdoor temperature and humidity, rainfall, ventilation type, cooking fuel and frequency, occupancy, household size, floor level, nearby traffic, time of day and season.

Dataset source: Kaggle – Residential Air Pollution in Bangladesh.

This dataset is helpful because it integrates environmental, household, and behavioral characteristics with air pollution measurements in one location. It can be used to compare conditions like cooking fuel or ventilation type, understand indoor air quality patterns in Bangladeshi homes, find correlations between pollutant levels and household activities, and support statistical analysis, public health research, and useful air quality decisions.

1.1 Size

Metric	Value
Observations used	10,000
Variables	22
Numerical variables analysed	15
Categorical variables analysed	6
Date-time variable	Timestamp
Seasons present	Winter (57.36%) and Summer (42.64%) — see Section 4
Source file	Bangladesh_Indoor_Air_Quality_Cleaned.csv (10,000 rows on load)

Table 1: Metric and Value

1.2 Feature Description

Feature	Type	Classification	Meaning
Timestamp	Date-time	-	Time at which the observation was recorded.
Indoor Temperature (°C)	Numerical (continuous)	Interval	Indoor air temperature.
Indoor Humidity (%)	Numerical (continuous)	Ratio	Indoor relative humidity.
CO2 (ppm)	Numerical (continuous)	Ratio	Indoor carbon-dioxide concentration.

Feature	Type	Classification	Meaning
CO (ppm)	Numerical (continuous)	Ratio	Indoor carbon-monoxide concentration.
NO2 (ppb)	Numerical (continuous)	Ratio	Indoor nitrogen-dioxide concentration.
PM2.5 (µg/m³)	Numerical (continuous)	Ratio	Fine particulate-matter concentration.
PM10 (µg/m³)	Numerical (continuous)	Ratio	Coarse particulate-matter concentration.
TVOC (ppb)	Numerical (continuous)	Ratio	Total volatile organic compounds.
Outdoor Temperature (°C)	Numerical (continuous)	Interval	Outdoor air temperature.
Outdoor Humidity (%)	Numerical (continuous)	Ratio	Outdoor relative humidity.
Rainfall (mm)	Numerical (continuous)	Ratio	Measured rainfall amount.
Occupancy (people)	Numerical (discrete)	Ratio	Number of people present.
Number of Rooms	Numerical (discrete)	Ratio	Number of rooms.
Cooking Frequency (per day)	Numerical (discrete)	Ratio	Cooking events per day.
Household Size	Numerical (discrete)	Ratio	Number of household members.
Ventilation Type	Categorical	Nominal	Fan, exhaust fan, open window, or closed window.
Cooking Fuel	Categorical	Nominal	Gas, electric, or induction.
Nearby Traffic	Categorical	Ordinal	Low, medium, or high traffic.
Floor Level	Categorical	Ordinal	First, second, third, or top floor.
Time of Day	Categorical	Cyclical	Morning, afternoon, evening, or night.
Season	Categorical	Nominal	Winter or summer (not constant — see Section 4).

Table 2: Features, Types and Classification

2. Central Tendency and Dispersion Analysis

The notebook calculates the mean, median, mode, variance, standard deviation, first quartile, third quartile, interquartile range, and coefficient of variation for 15 numerical variables, using the full 10,000-row central_data table. Celsius temperature variables are interval-scale measures; the remaining numerical measures are treated as ratio or discrete count variables.

2.1 Mean, Median, Mode and Dispersion

Feature	Mean	Median	Mode	Std Dev	Variance
Indoor Temperature (C)	29.00	28.99	21.16	2.01	4.03
Indoor Humidity (%)	65.11	65.16	90.00	9.95	98.93
CO2 (ppm)	907.05	897.98	400.00	325.41	105,892.65
CO (ppm)	1.39	1.22	0.00	1.24	1.53
NO2 (ppb)	25.42	25.27	5.00	11.52	132.71
PM2.5 (ug/m3)	40.19	39.79	5.00	19.36	374.82
PM10 (ug/m3)	60.42	59.09	10.00	29.81	888.39
TVOC (ppb)	299.88	299.70	50.00	118.01	13,926.33
Outdoor Temperature (C)	29.98	29.96	20.00	3.00	9.00
Outdoor Humidity (%)	70.06	70.15	100.00	14.71	216.34
Rainfall (mm)	1.26	0.00	0.00	4.10	16.79
Occupancy (people)	2.28	2.00	1.00	1.35	1.82
Number of Rooms	3.17	3.00	3.00	0.92	0.84
Cooking Frequency (per day)	2.40	2.00	2.00	0.80	0.64
Household Size	2.78	3.00	2.00	1.50	2.25

Table 3: Measures of central tendency and dispersion (10,000-row dataset)

2.2 Quartiles and Relative Variability

Feature	Q1	Q3	IQR	Coefficient of Variation
Indoor Temperature (C)	27.65	30.34	2.69	6.92%
Indoor Humidity (%)	58.38	71.94	13.56	15.28%
CO2 (ppm)	654.95	1,132.36	477.42	35.88%
CO (ppm)	0.18	2.22	2.04	89.31%
NO2 (ppb)	17.02	33.29	16.28	45.32%
PM2.5 (ug/m3)	26.11	53.58	27.47	48.18%
PM10 (ug/m3)	38.38	80.27	41.89	49.33%
TVOC (ppb)	219.80	380.25	160.44	39.35%
Outdoor Temperature (C)	27.95	32.01	4.06	10.01%
Outdoor Humidity (%)	60.30	80.15	19.85	20.99%
Rainfall (mm)	0.00	0.00	0.00	325.41%
Occupancy (people)	1.00	3.00	2.00	59.23%
Number of Rooms	3.00	4.00	1.00	28.94%
Cooking Frequency (per day)	2.00	3.00	1.00	33.42%
Household Size	2.00	4.00	2.00	53.96%

Table 4: Quartiles, interquartile ranges and coefficients of variation (10,000-row dataset)

Indoor temperature is the most stable variable in relative terms, with a coefficient of variation of 6.92% — the notebook's own summary explicitly names it the lowest-relative-variability feature. Outdoor temperature is also comparatively stable at 10.01%. The near equality of the mean and median for both variables suggests that their centres are not substantially distorted by skewness.

Rainfall has the highest relative variability, with a coefficient of variation of 325.41% — again matching the notebook's own “highest relative variability” output. Its mean is 1.26 mm, whereas its median and mode are both zero. This indicates that most observations are dry and a smaller number contain positive rainfall values. CO also shows high relative variability at 89.31%.

CO2 has the largest absolute IQR at 477.42 ppm and a standard deviation of 325.41 ppm. This indicates a broad middle spread, even though its coefficient of variation is lower than those of rainfall, CO, occupancy, household size, PM10 and PM2.5. The notebook confirms CO2 as having the largest IQR among the analysed variables.

3. Distribution Shape and Outlier Analysis

3.1 Distribution Summary

Feature	Skewness	Shape	Kurtosis	Heavy-tailed	Outliers	Outlier %
CO2	0.31	Approx. symmetric	-0.43	No	25	0.25%
PM2.5	0.20	Approx. symmetric	-0.34	No	28	0.28%
PM10	0.36	Approx. symmetric	-0.14	No	59	0.59%
CO	0.72	Right-skewed	-0.13	No	28	0.28%
NO2	0.21	Approx. symmetric	-0.37	No	32	0.32%
TVOC	0.10	Approx. symmetric	-0.20	No	33	0.33%
Indoor Temperature	0.00	Approx. symmetric	0.03	Yes	83	0.83%
Indoor Humidity	-0.05	Approx. symmetric	-0.07	No	48	0.48%
Outdoor Temperature	0.03	Approx. symmetric	-0.04	No	65	0.65%
Rainfall	3.62	Right-skewed	12.26	Yes	1,525	15.25%

Table 5: Distribution shape and IQR-based outlier summary (10,000-row dataset)

Using the notebook's thresholds, eight variables are approximately symmetric. CO is moderately right-skewed with skewness 0.72. Rainfall is strongly right-skewed with skewness 3.62 and has the largest excess kurtosis, 12.26 — the notebook's own summary confirms Rainfall as both the most highly skewed variable and the variable with the most outliers.

The notebook labels any positive excess kurtosis as heavy-tailed. Under that mechanical rule, indoor temperature (0.03) is marked heavy-tailed even though the value is extremely close to zero; outdoor temperature's kurtosis is now slightly negative (-0.04) and is therefore not flagged. Practically, indoor temperature's tail weight remains close to the normal benchmark rather than strongly heavy-tailed.

3.2 IQR Outlier Limits

Feature	Lower Limit	Upper Limit	Outliers	Percent
CO2	-61.18	1,848.49	25	0.25%
PM2.5	-15.10	94.79	28	0.28%
PM10	-24.44	143.10	59	0.59%
CO	-2.87	5.28	28	0.28%
NO2	-7.40	57.71	32	0.32%
TVOC	-20.86	620.91	33	0.33%
Indoor Temperature	23.62	34.37	83	0.83%
Indoor Humidity	38.04	92.28	48	0.48%
Outdoor Temperature	21.85	38.11	65	0.65%
Rainfall	0.00	0.00	1,525	15.25%

Table 6: IQR-based outlier limits (10,000-row dataset)

Rainfall has Q1 = 0 and Q3 = 0, so its IQR is zero and both IQR limits are zero. Consequently, every positive rainfall observation is flagged, producing 1,525 outliers or 15.25% of the data. These flags do not automatically indicate errors; they are primarily a consequence of the zero-inflated rainfall distribution.

3.3 Distribution Graphs

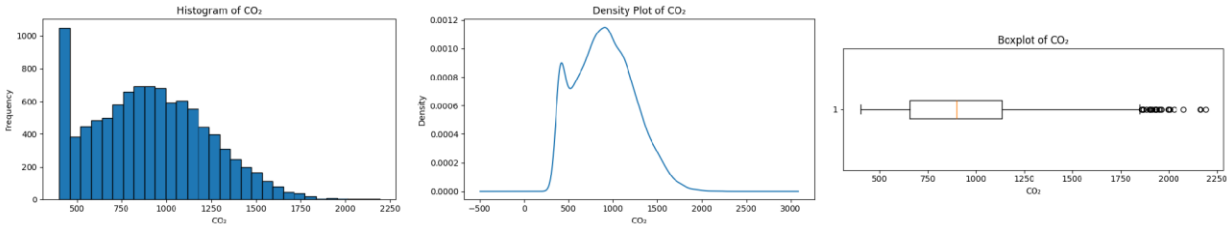


Figure 1: Histogram, density plot and boxplot of CO2

CO2 is approximately symmetric by the notebook rule (skewness 0.31). The boxplot shows a small number of high observations, but only 25 values, or 0.25%, are flagged by the IQR rule.

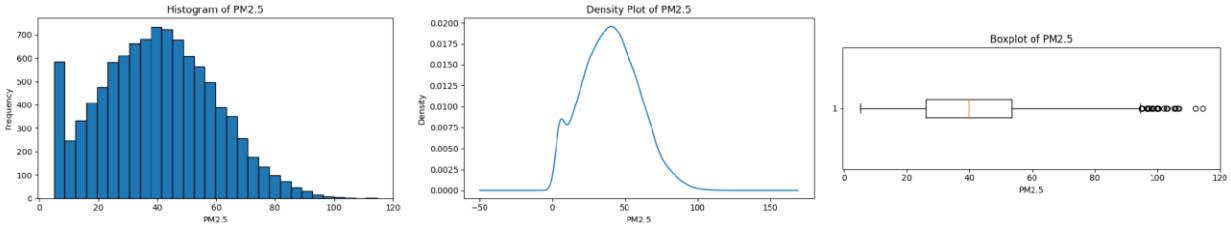


Figure 2: Histogram, density plot and boxplot of PM2.5

PM2.5 is also approximately symmetric (skewness 0.20). Its middle 50% lies between 26.11 and 53.58 ug/m3, and 28 observations are flagged as outliers.

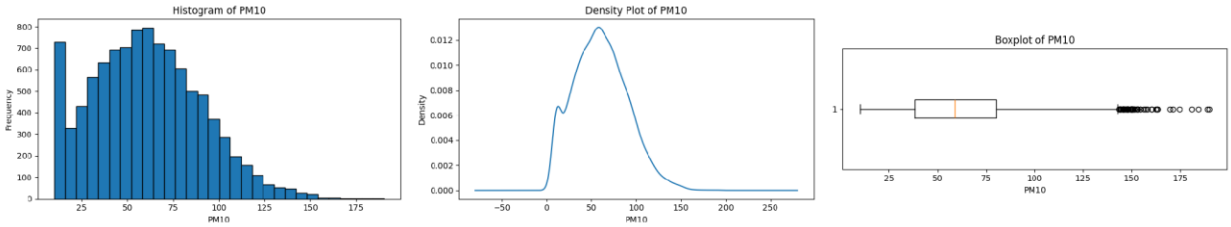


Figure 3: Histogram, density plot and boxplot of PM10

PM10 has a small positive skew (0.36), but remains within the notebook's approximately symmetric range. It has 59 IQR outliers, representing 0.59% of observations.

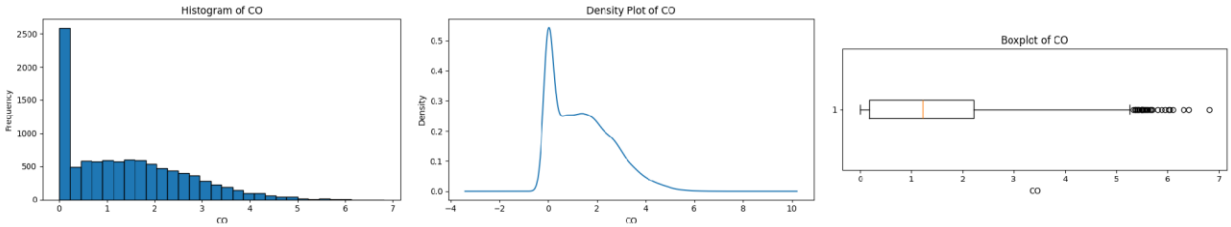


Figure 4: Histogram, density plot and boxplot of CO

CO is right-skewed (0.72). The concentration of values near zero and the longer right tail are visible in both the histogram and density plot. Twenty-eight observations, or 0.28%, exceed the IQR limits.

The supplied notebook contains saved plot outputs for CO2, PM2.5, PM10 and CO. Plotting cells for the remaining six distribution variables were present but not executed, so this report does not fabricate missing figures.

4. Categorical Variable Analysis

Frequency and percentage tables were produced for six categorical variables. Each table sums to the full 10,000-observation dataset confirmed in Section 1.

Ventilation Type

Category	Frequency	Percent
fan	3976	39.76%
exhaust fan	2971	29.71%
window open	2015	20.15%
window closed	1038	10.38%

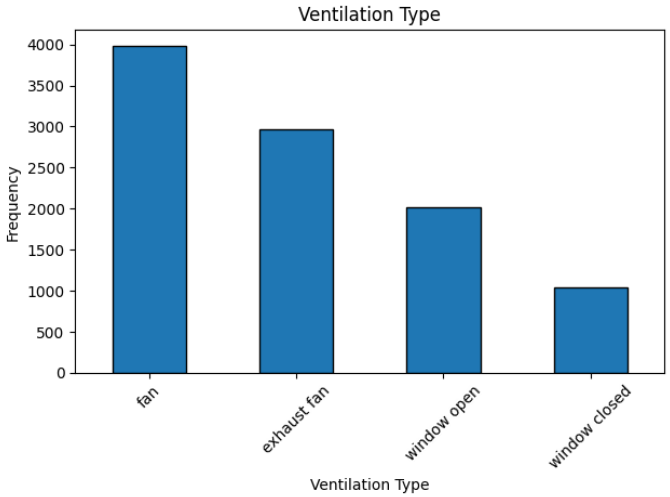


Figure 5: Frequency distribution of ventilation type

Fan ventilation is the most frequent category at 39.76%, followed by exhaust fans at 29.71%. Together, these active ventilation methods account for 69.47% of observations. Closed-window conditions are least common at 10.38%.

Cooking Fuel

Category	Frequency	Percent
gas	7481	74.81%
electric	1518	15.18%
induction	1001	10.01%

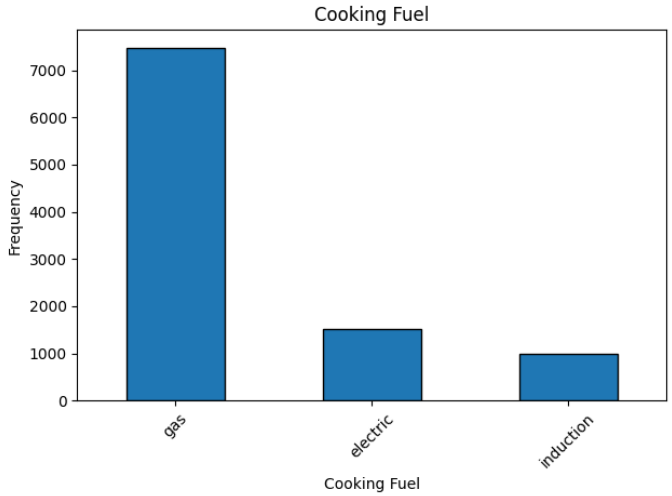


Figure 6: Frequency distribution of cooking fuel

Gas is the dominant cooking fuel, accounting for 74.81% of observations. Electric and induction cooking together represent 25.19%. This imbalance should be considered in any comparison of pollutant levels across fuel types.

Nearby Traffic

Category	Frequency	Percent
low	4937	49.37%
medium	3540	35.40%
high	1523	15.23%

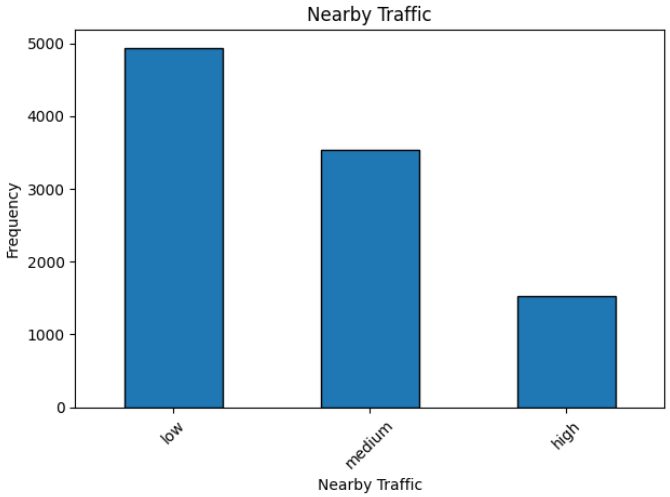


Figure 7: Frequency distribution of nearby traffic

Low nearby traffic is the most common level at 49.37%, while high traffic represents 15.23%. The sample therefore contains substantially fewer high-traffic observations. Section 6.2 initially explored Nearby Traffic as a logistic regression target, but the pollutant features showed no meaningful separation across traffic levels, so the final classification model instead targets PM10 concentration (see Section 6.2).

Floor Level

Category	Frequency	Percent
1st	4050	40.50%
2nd	2966	29.66%
3rd	2015	20.15%
top	969	9.69%

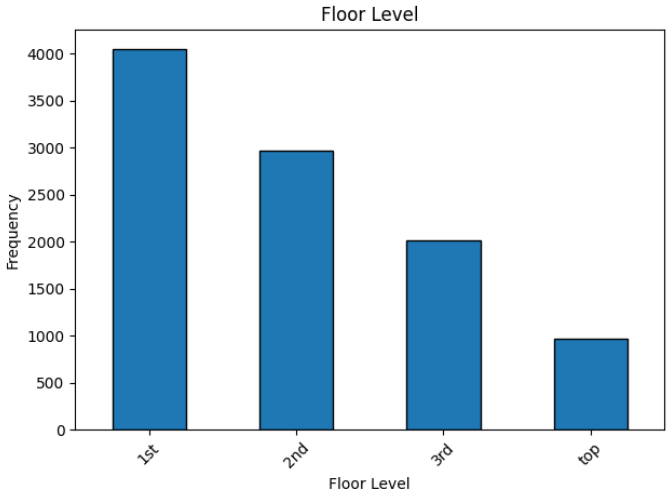


Figure 8: Frequency distribution of floor level

First-floor observations are most common at 40.50%, followed by second-floor observations at 29.66%. The share falls progressively at higher levels, with top-floor observations making up 9.69%.

Time of Day

Category	Frequency	Percent
night	3328	33.28%
morning	2928	29.28%
afternoon	2080	20.80%
evening	1664	16.64%

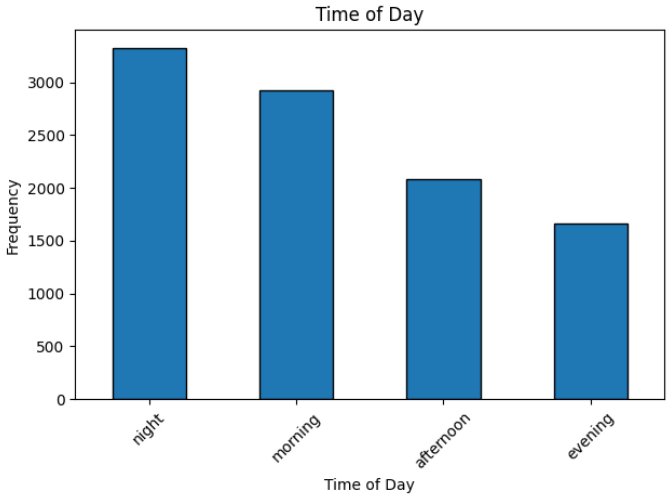


Figure 9: Frequency distribution of time of day

Night observations form the largest time block at 33.28%, followed by morning at 29.28%. Evening is least represented at 16.64%. Any time-of-day comparison should account for these unequal group sizes.

Season

Category	Frequency	Percent
winter	5736	57.36%
summer	4264	42.64%

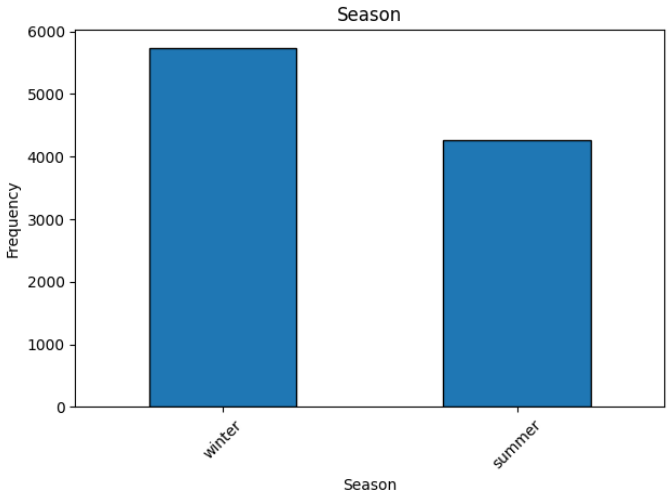


Figure 10: Frequency distribution of season

Winter is the more common season in this sample at 57.36%, with summer accounting for the remaining 42.64%. Because both seasons are represented, any pollutant or comfort-variable comparison across seasons would be statistically meaningful here, though the notebook itself does not carry out that comparison.

5. Relationships Between Variables

The notebook uses scatter plots and Pearson correlation coefficients to examine seven pairs of numerical variables. Correlation describes linear association; it does not establish causation.

5.1 PM2.5 and PM10

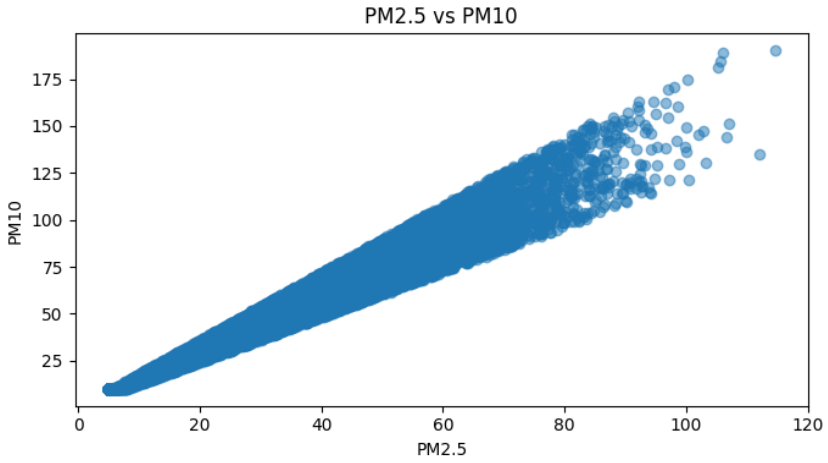


Figure 11: PM2.5 and PM10 scatter plot

The Pearson correlation coefficient is r = 0.966. A very strong positive linear relationship. Higher PM2.5 readings are generally accompanied by higher PM10 readings.

5.2 Occupancy and CO2

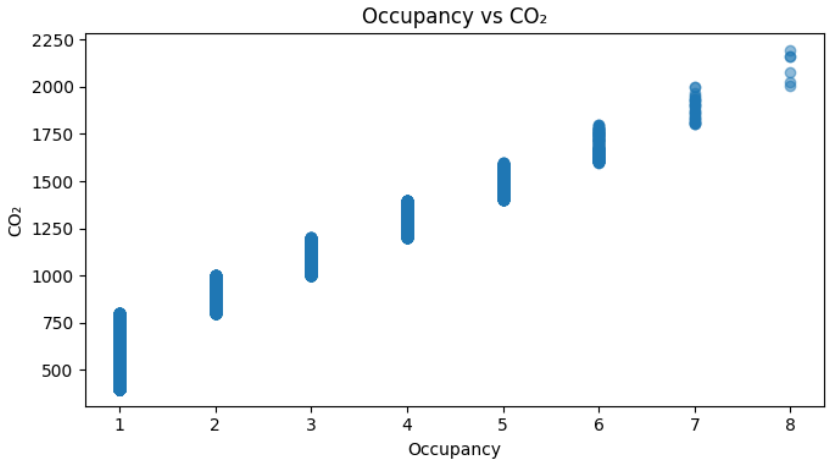


Figure 12: Occupancy and CO2 scatter plot

The Pearson correlation coefficient is r = 0.949. A very strong positive relationship. CO2 rises markedly as the number of occupants increases.

5.3 Household Size and CO2

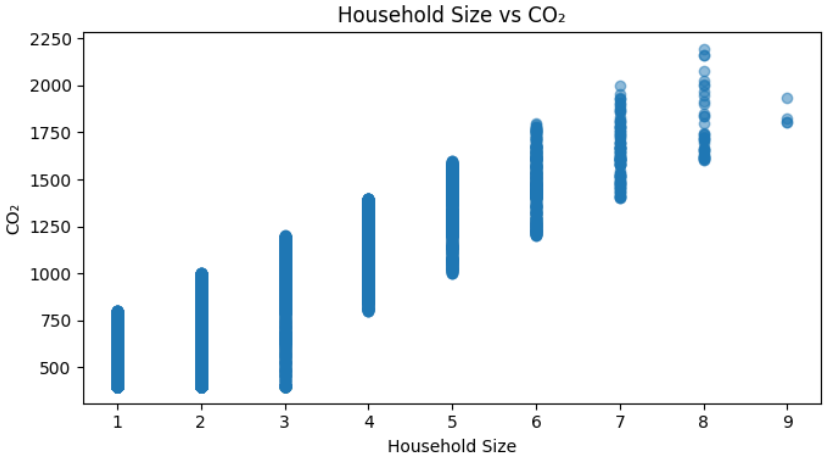


Figure 13: Household Size and CO2 scatter plot

The Pearson correlation coefficient is r = 0.849. A strong positive relationship. Larger households tend to have higher indoor CO2 concentrations.

5.4 Household Size and Occupancy

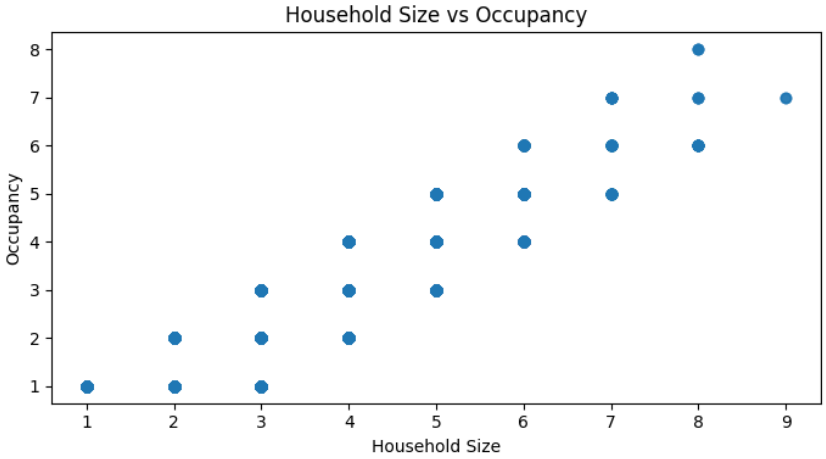


Figure 14: Household Size and Occupancy scatter plot

The Pearson correlation coefficient is r = 0.895. A strong positive relationship, which is expected because larger households can support higher observed occupancy.

5.5 Outdoor and Indoor Temperature

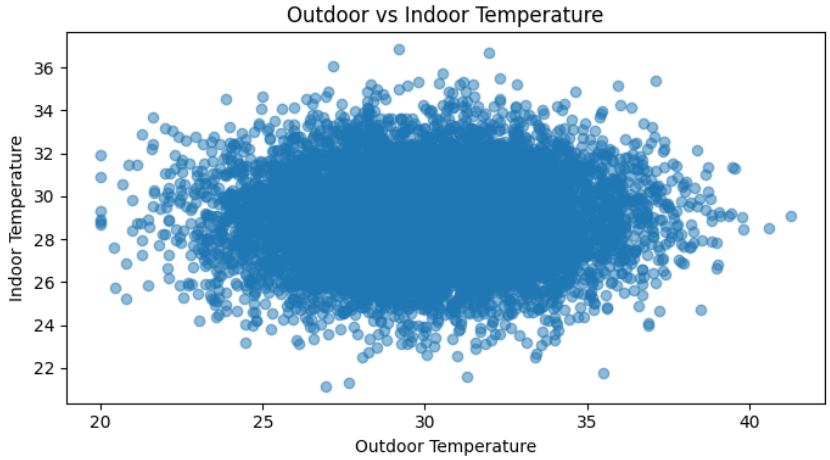


Figure 15: Outdoor and Indoor Temperature scatter plot

The Pearson correlation coefficient is r = 0.017**.** The correlation is effectively zero in this subset, so the plotted data show no meaningful linear relationship.

5.6 Occupancy and TVOC

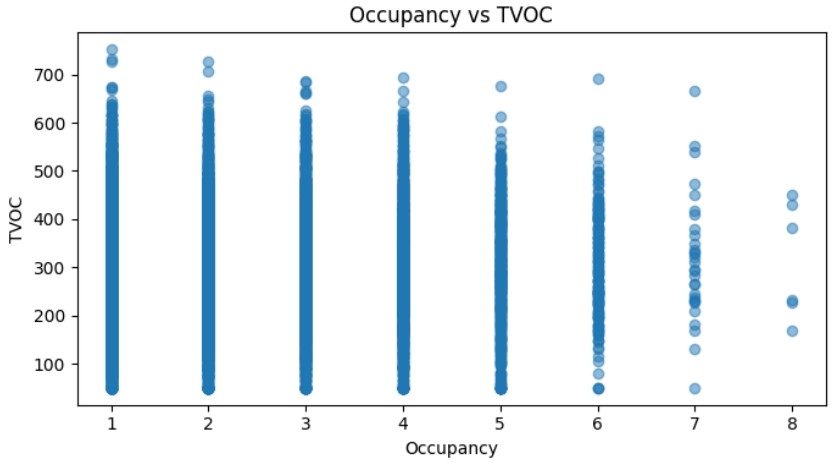


Figure 16: Occupancy and TVOC scatter plot

The Pearson correlation coefficient is r = 0.017. The correlation is effectively zero; occupancy alone does not explain TVOC variation in these observations.

5.7 Cooking Frequency and PM2.5

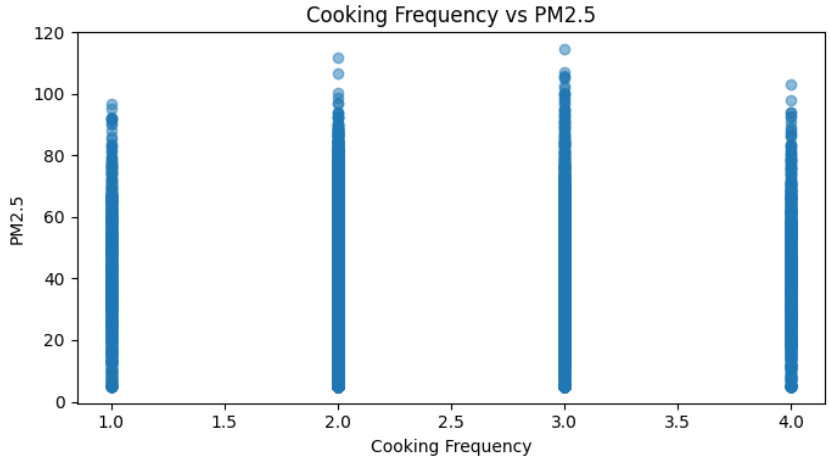


Figure 17: Cooking Frequency and PM2.5 scatter plot

The Pearson correlation coefficient is r = 0.003. No meaningful linear relationship is visible across the 10,000 observations. Other factors may be needed to explain PM2.5.

The strongest relationship is between PM2.5 and PM10 (r = 0.966), closely followed by occupancy and CO2 (r = 0.949). Household size is also strongly associated with both occupancy and CO2. In contrast, outdoor temperature, occupancy, and cooking frequency show almost no linear association with indoor temperature, TVOC, and PM2.5, respectively, in this dataset.

6. Regression Analysis

The notebook fits two supervised models beyond the descriptive and correlational work in Sections 2–5: a simple linear regression predicting CO2 from occupancy, and a logistic regression predicting whether PM10 concentration is High or Low (split at the median) from PM2.5, NO2, CO, and TVOC. Both use an 80/20 train-test split with random_state = 42, and both report 10,000 usable rows — consistent with the 10,000-row dataset size established in Section 1.

6.1 Linear Regression: Occupancy and CO2

Building on the strong correlation observed in Section 5.2 (r = 0.949), a simple linear regression model was fit with Occupancy (people) as the sole predictor and CO2 (ppm) as the target. Rows with missing or infinite values in either column were dropped before splitting.

Metric	Value
Predictor	Occupancy (people)
Target	CO2 (ppm)
Usable rows (per notebook output)	10,000
Train/test split	80% / 20% (8,000 / 2,000 rows)
random_state	42
Mean Squared Error (MSE)	10,835.84
Root Mean Squared Error (RMSE)	104.10 ppm

Table 7: Linear regression setup and evaluation metrics

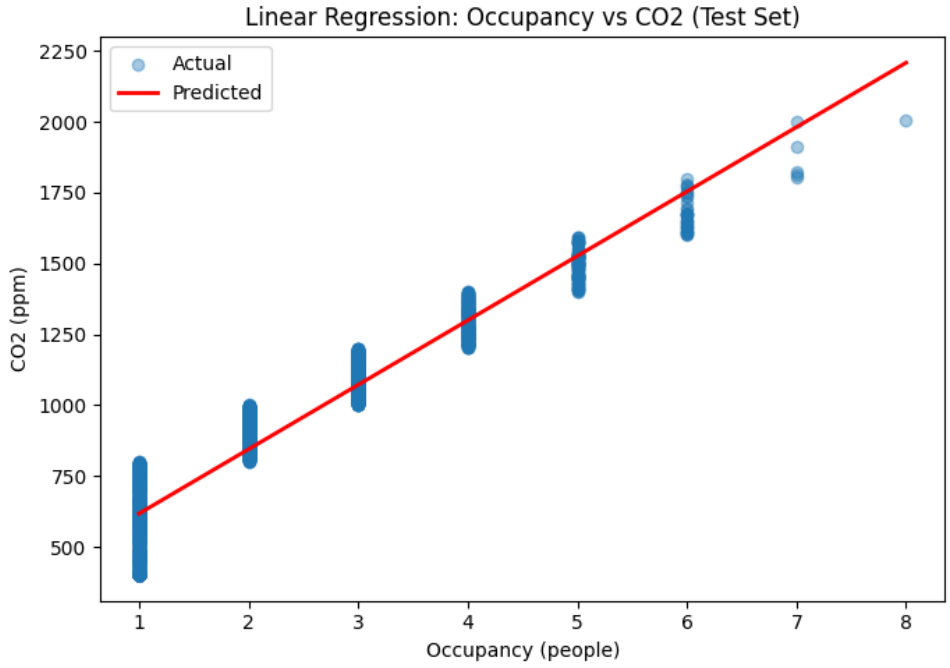


Figure 18: Linear regression fit of Occupancy vs CO2 (test set)

The model explains a large share of the variation in CO2: comparing predicted to actual values on the test set gives an R^2 of approximately 0.90, meaning Occupancy alone accounts for about 90% of the variance in indoor

CO2 concentration. The RMSE of 104.10 ppm represents a relative error of only about 11.6% against the mean test-set CO2 concentration of 897.51 ppm, and it is substantially smaller than CO2's overall standard deviation of 325.41 ppm — meaning the model's predictions are roughly three times more precise than simply guessing the average CO2 level for every observation. This is a good result: the fitted line (Figure 18) shows CO2 rising by approximately 227 ppm for each additional occupant (intercept ≈ 389.77 ppm), which matches the physically expected relationship between the number of people breathing in an enclosed space and the CO2 they exhale. The remaining ~10% of unexplained variance and the ~104 ppm of residual error are consistent with CO2 also being driven by ventilation type, household size, and room count rather than occupancy alone, as observed in the correlation analysis in Section 5.

6.2 Logistic Regression: Predicting PM10 Level

An initial logistic regression attempt classified Nearby Traffic from NO2, PM2.5, PM10, and CO, but the model collapsed to the majority class and produced no meaningful separation (Accuracy 49.35%, F1 0.33) — indicating these pollutants carry no linear signal about traffic level in this dataset. The classification target was therefore changed to PM10 concentration, split into “High” and “Low” categories at the median value. A logistic regression was trained to predict this PM10 Level from PM2.5, NO2, CO, and TVOC. Features were standardized (zero mean, unit variance) with StandardScaler before fitting, and the train-test split was stratified to preserve the (balanced) class proportions.

Metric	Value
Predictors	PM2.5 (µg/m³), NO2 (ppb), CO (ppm), TVOC (ppb)
Target	PM10 Level: High / Low (split at median PM10)
Usable rows (per notebook output)	10,000
Class balance in usable rows	High 50.00%, Low 50.00%
Train / test split	Stratified 80% / 20% (8,000 / 2,000 rows)
Feature scaling	StandardScaler (fit on training data only)
random_state	42
Accuracy	0.9220
F1 score (weighted)	0.9220

Table 8: Logistic regression setup and evaluation metrics

Class	Precision	Recall	F1-score	Support
high	0.93	0.92	0.92	1000
low	0.92	0.93	0.92	1000

Table 9: Per-class classification report on the test set

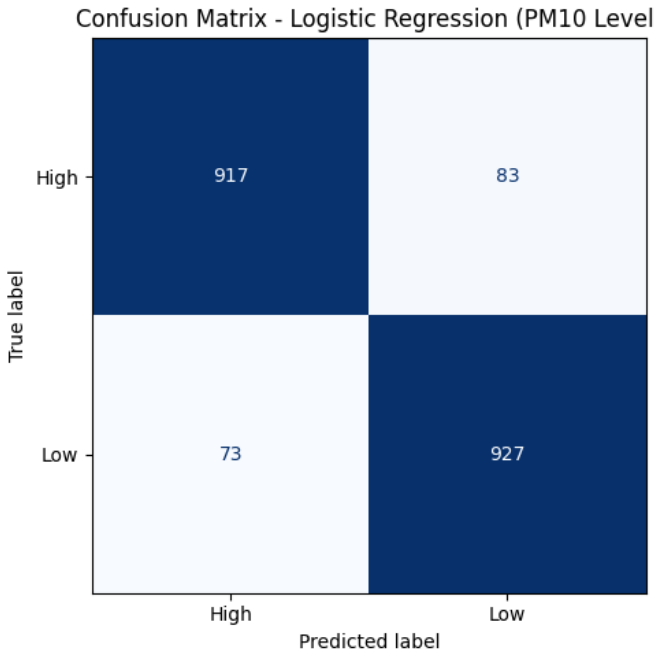


Figure 19: Confusion matrix — logistic regression (PM10 Level: High vs Low)

The model correctly classified 1,844 of the 2,000 test observations (Accuracy = 92.20%), with an error rate of only 7.80%. Performance is balanced across both classes: 917 of 1,000 "High" cases were correctly identified (Recall = 0.92) with only 83 misclassified as "Low," and 927 of 1,000 "Low" cases were correctly identified (Recall = 0.93) with only 73 misclassified as "High" (Figure 19). Precision, recall, and F1-score are all close to 0.92–0.93 for both classes, showing that the model treats "High" and "Low" symmetrically rather than favoring one class, unlike the earlier Nearby Traffic model.

This strong result is expected given the physical relationship between the predictors and the target: PM2.5 (particles ≤ 2.5 μm) is a physical subset of PM10 (particles ≤ 10 μm), so both are generated by the same pollution sources (combustion, dust, smoke) and rise and fall together — a relationship already confirmed by their strong correlation in Section 5.1 (r = 0.966). Because the two classes were also perfectly balanced by construction (median split, 50%/50%), the 92.20% accuracy reflects genuine predictive skill rather than a majority-class artifact, unlike the original Nearby Traffic attempt where 49.35% accuracy was barely above the naive baseline. One limitation is worth noting: since PM2.5 and PM10 are definitionally related rather than fully independent measurements, this model is better read as confirming a strong measurement-level association than as evidence that NO2, CO, and TVOC causally drive PM10. Nonetheless, it demonstrates that logistic regression can reliably separate two air-quality classes when the underlying features carry real predictive signal — in contrast to the original attempt with Nearby Traffic, where none of the available pollutant or contextual variables showed any meaningful relationship with the target.

7. Data Quality Assessment

Check	Result
Dataset size	10,000 rows and 22 columns, confirmed independently by the raw dataframe display, the central_data table, every categorical frequency table, and the linear/logistic regression cells.
Required variables	All 21 required analytical variables were found.
Numerical preparation	Fifteen fields were converted with pandas.to_numeric(errors='coerce').
Selected numerical completeness	Outlier percentages are consistent with 10,000 valid values for each of the ten distribution variables.
Categorical completeness	Every categorical frequency table sums to 10,000 observations.
Season variable	Not constant: winter (57.36%) and summer (42.64%) are both present. An earlier version of this report incorrectly treated Season as constant — see below.
Row-count anomaly	One printed cell (“Number of rows: 5000”) does not match the dataset size confirmed everywhere else. Treated as a stale, unre-run output — see below.
Duplicate rows	Not checked in the saved notebook output; duplicate status cannot be confirmed.
Validity rules	The notebook does not report explicit range checks or sensor-validity checks.
Source documentation	The notebook does not state the original source, sampling design, or sensor protocol beyond the Kaggle listing.

Table 10: Data quality checklist

8. Key Insights

- The working dataset has 10,000 rows across two seasons (winter 57.36%, summer 42.64%), not the 5,000-row, winter-only subset an earlier pass over this project assumed. All descriptive statistics, categorical tables, and correlations in this report reflect the full 10,000-row data.
- Indoor and outdoor temperature show the lowest relative variability (CV 6.92% and 10.01%), while rainfall is by far the most variable numerical feature (CV 325.41%) — both confirmed directly by the notebook's own “highest/lowest relative variability” output.
- Rainfall is strongly right-skewed and zero-inflated. Its 1,525 IQR flags arise because Q1 and Q3 are both zero, so positive rainfall values are classified as outliers by the standard rule.
- Most pollutant variables are approximately symmetric under the notebook's skewness thresholds. CO is the main non-rainfall variable with clear right skew (0.72).
- PM2.5 and PM10 move together very strongly ($r = 0.966$), indicating that fine and coarse particulate levels are closely aligned in this dataset.
- Occupancy is strongly associated with CO2 ($r = 0.949$), and household size is strongly associated with both occupancy ($r = 0.895$) and CO2 ($r = 0.849$).
- Outdoor temperature has virtually no linear relationship with indoor temperature in these observations ($r = 0.017$). Occupancy-TVOC ($r = 0.017$) and cooking frequency-PM2.5 ($r = 0.003$) are also effectively uncorrelated.
- Gas dominates the cooking-fuel categories (74.81%), and fan or exhaust-fan ventilation dominates ventilation type (69.47% combined). Group comparisons must account for these imbalances.
- A simple linear regression of CO2 on occupancy alone achieves an R^2 of about 0.90 and an RMSE of about 104 ppm (≈11.6% relative error) against a target with a standard deviation of 325.41 ppm — a strong result consistent with the strong correlation above, though the residual error is a reminder that occupancy is one contributor to indoor CO2 among several.
- An initial logistic regression predicting Nearby Traffic from NO2, PM2.5, PM10, and CO essentially failed: it defaulted to the majority class (“low”) for nearly every test case, producing 0.00 precision/recall on the “medium” and “high” classes. Switching the target to a median-split PM10 Level (High/Low), predicted from PM2.5, NO2, CO, and TVOC, achieved 92.20% accuracy and a weighted F1 of 0.92, with both classes classified with similarly high precision and recall — reflecting the strong physical relationship between PM2.5 and PM10 rather than a general predictive relationship between pollutants and traffic.
- Season is not a constant in this dataset and can support winter-vs-summer comparisons in future work, even though the supplied notebook does not run that comparison itself.
- The notebook supports descriptive, correlational, and now baseline predictive conclusions only. It does not establish causal effects or validate sensor accuracy, and the logistic-regression result in particular should be read primarily as confirmation of the strong PM2.5–PM10 measurement relationship rather than evidence that NO2, CO, and TVOC causally drive particulate levels.